Voevodsky's Lectures on Motivic Cohomology 2000/2001

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Voevodsky's Lectures on Motivic Cohomology

Abstract

These lectures cover several important topics of motivic homotopy theory which have not been covered elsewhere. These topics include the definition of equivariant motivic homotopy categories, the definition and basic properties of solid and ind-solid sheaves and the proof of the basic properties of the operations of twisted powers and group quotients relative to the A -equivalences between ind-solid sheaves.

1 Introduction

The lectures which provided the source for these notes covered several different topics which are related to each other but which do not in any reasonable sense form a coherent whole. As a result, this text is a collection of four parts which refer to each other but otherwise are independent. In the first part we introduce the motivic homotopy category and connect it with the motivic cohomology theory discussed in [7]. The exposition is a little unusual because we wanted to avoid any references to model structures and still prove the main theorem 2. We were able to do it modulo 6 where we had to refer to the next part. The second part is about we the motivic homotopy category of -schemes where is a finite flat group scheme with respect to an equivariant analog of the Nisnevich topology. Our main result is a description of the class of A -equivalences (formerly called A -weak equivalences) given in Theorem 4 (also in Theorem 5). For the trivial group we get a new description of the A -equivalences in the non equivariant setting. In the third part we define a class of sheaves on -schemes which we call solid sheaves. It contains all representable sheaves and quotients of representable sheaves by subsheaves corresponding to open subschemes. In particular the Thom P. Deligne e-mail: deligne@math.ias.edu N.A. Baas et al. (eds.), Algebraic Topology, Abel Symposia 4,

355 356 P. Deligne spaces of vector bundles are solid sheaves. The key property of solid sheaves can be expressed by saying that any right exact functor which takes open embeddings to monomorphisms is left exact on solid sheaves. A more precise statement is

Theorem 1 (6)**.** *In the fourth part we study two functors. One is the extension to pointed sheaves of the functor from -schemes to schemes which takes to . The other one is extension to pointed sheaves of the functor which takes to] where is a finite flat -scheme. We show that both functors take solid sheaves to solid sheaves and preserve local and A -equivalences between termwise (ind-)solid sheaves. The material of all the parts of these notes but the first one was originally developed with one particular goal in mind – to extend non-additive functors, such as the symmetric product, from schemes to the motivic homotopy category. More precisely, we were interested in functors given by*

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<] where is a finite flat group scheme, ? is a finite flat -scheme and any -scheme of finite type. The equivariant motivic homotopy category was introduced to represent < as a composition]]] and solid sheaves as a natural class of sheaves on which the derived functor L< coincides with < . In the present form these notes are the result of an interactive process which involved

all listeners of the lectures. A very special role was played by Pierre Deligne. The text as it is now was completely written by him. He also cleared up a lot of messy parts and simplified the arguments in several important places. Princeton, NJ, 2001. Vladimir Voevodsky

2 Motivic Cohomology and Motivic Homotopy Category

We will recall first some of last year results (see [7]).

2.1 Last Year

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1.1 We work over a field which sometimes will have to be assumed to be perfect. The schemes over we consider will usually be assumes separated and smooth of finite type over . We note their category. Three Grothendieck topologies on will be useful: Zariski, Nisnevich and etale. For each of these topologies a

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357 ! /

/ / 1.2 The definition of the motivic cohomology groups of smooth over ! has the sheaf on amounts to the data for smooth over of a sheaf - on the small site [#) (resp. . , 0) of the open subsets of (resp. of etale), with - functorial in : a map induces X -.

following form:

(a) One defines for each Z a complex of presheaves of abelian groups Z on . It is in fact a complex of sheaves for the etale topology, hence a fortiori for the Nisnevich and Zariski topology. For any abelian group the same applies to Z . (b) The motivic cohomology groups of with coefficients in are the hypercohomology groups of the , in the Nisnevich topology:

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% % H . %

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For Z we will write simply . Motivic cohomology has the following properties:

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1. The complex Z is zero for . For any it lives in cohomological degree . As a complex of Nisnevich sheaves it is quasi-isomorphic to Z for and to G for . J 2. for any .

3. For any in one has

,L! ! , (! = , where = , is the ,-th higher Chow group of cycles of codimension . 4. In the etale topology, for $ prime to the characteristic of ! , the complex Z$

is quasi-isomorphic to D , giving for the etale analog of the formula 0 Z$ H 0 Z$ 0 D 1.3 The category (=AB! is the category with objects separated schemes smooth of finite type over ! , for which a morphism C is a cycle C $ C on each of whose irreducible components C is finite over and projects onto a connected component of . A morphism C can be thought of as a finitely valued map from to . For , with residue field !, it defines a zero cycle C on , and the assumption made on C implies that the degree of this 0-cycle is locally constant on . A morphism of schemes defines a morphism in (=AB!: the graph of . This graph construction defines a faithful functor from (! to the additive category (=AB!. A presheaf with transfers is a contravariant additive functor from the category (=AB! to the category of abelian groups. The embedding of (! in

358 P. Deligne (=AB! allows us to view a presheaf with transfers as a presheaf on (! endowed with an extra structure. A sheaf with transfers (for a given topology on (!, usually the Nisnevich topology) is a presheaf with transfers which, as a presheaf on (! , is a sheaf. The Nisnevich and the etale topologies have the virtue that if / is a presheaf with transfers, the associated sheaf 7/ carries a structure of a sheaf with transfers. This structure is uniquely determined by / 7/ being a morphism of presheaves with transfers. For any sheaf with transfers

, one has

7/ Hom/

Hom

(Hom of presheaves with transfers). All of this fails for the Zariski topology. The complexes Z (or) start life as complexes of sheaves with transfers.

% 1.4 A presheaf / on (! is called homotopy invariant if / / A . As the point of A defines a section of the projection of A to , for any presheaf of abelian groups / , / is naturally a direct factor of / A ; it follows that the condition "homotopy invariant' is stable by kernels, cokernels and extensions of presheaves. The following construction is a derived version of the left adjoint to the inclusion

A(AA,. '$v7B'7$,BL& L7vL& 7MM ,BL& L7vL& (a) For a finite set, let % be the affine space freely spanned (in the sense of barycentric calculus) by . Over C, or R, % contains the standard topological simplex spanned by . The schemes 0 % $ form a cosimplicial scheme. (b) For / a presheaf, = / (the "singular complex of / ') is the simplicial presheaf = / / 0 . Arguments imitated from topology show that for / a presheaf of abelian groups, the cohomology presheaves of the complex = / , obtained from = / by taking alternating sum of the face maps, are homotopy invariant. If / has transfers so do the = / and hence the = / . A basic theorem proved last year is:

Theorem 2 (1). *Let / be a homotopy invariant presheaf with transfers over a perfect field with the associated Nisnevich sheaf 7. / . Then the presheaves with transfers . 7. / are homotopy invariant as well.*

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The particular case of this theorem for claims the homotopy invariance of the sheaf with transfers . . Last year, the equivalence of a number of definitions of Z was proven. Equivalence means: a construction of an isomorphism in a suitable derived category, implying an isomorphism for the corresponding motivic cohomology groups. For our present purpose the most convenient definition is as follows.

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Let Z) be the sheaf with transfers represented by (on the category (=AB!). We set ' for J Z) A Z) A for and Z = .

2.2 Motivic Homotopy Category

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The motivic homotopy category (pointed A -homotopy category of), A for a finite dimensional noetherian scheme, will be the category deduced from a category of simplicial sheaves by two successive localizations.1 One starts with the category of schemes smooth over , with the Nisnevich topology, and the category of pointed simplicial sheaves on . For any site S (for instance), there is a notion of local equivalence of (pointed) . simplicial sheaves. It proceeds as follows:

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(a) A sheaf defines a simplicial sheaf maps the identities. The functor

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Hom
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Hom
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with all and all simplicial has a left adjoint :
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The sheaf can be described as the equalizer of sheaf associated to the presheaf
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/ / , as well as the
/ The same holds in the pointed context. We will often write simply for . is a simplicial sheaf, and u a section of over , one also disposes of (b) If sheaves u over : the sheaves associated to the presheaves
/
/
/
6 / 6 u /
(c) A morphism is a local equivalence, if it induces an isomorphism on as well as, for any local section u of , an isomorphism on all . This applies also to pointed simplicial sheaves: one just forgets the marked point.
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A ((
One defines as the category derived from the category of pointed sim plicial sheaves on . by formally inverting local equivalences. Until made more concrete, this definition could lead to set-theoretic difficulties, which we leave the reader to solve in its preferred way. 1
In the Appendix we have assembled the properties of “localization’ to be used in this talk and in the next.
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For a pointed sheaf on , Proposition 7 applies to tion by local equivalences: one has
and to the localiza-
/ Hom/ Hom / (0.1)

Definition 3 (1)**.** An object of A (is called A -local if for any simplicial sheaf , one has Hom*
Hom*
Hom* A A
At the right hand side, A means that in the product, A is contracted to a point, the new base point.

Proposition 4 (1)**.** *For a pointed sheaf on if and only if is homotopy invariant.*
(, the simplicial sheaf
is A -local
A, so that by (0.1) “A-local’ means that

Proof. We have A for any pointed sheaf , one has
A A A morphism can be viewed as the data, for each . , of . , functorial in and marked point going to marked point. A morphism can similarly be described as data for . of . A . A Homotopy invariance hence implies A -locality. The converse is checked by taking for the disjoint sum of a representable sheaf and the base point.

□

Definition 5 (2)**.** (1) A morphism in A(is an A-equivalence if for any A -local , one has in A (Hom

Hom
Hom (2) The category AA (is deduced from A (by formally inverting Hom
A -equivalences.

Remark 6 (1)**.** If a morphism in A (becomes an isomorphism in AA (it is an A-equivalence. Indeed, if in A(is A-local, an application of Proposition 7 shows that for any , Hom* Hom* A
A (
AA (which
is bijective. If in has an image in
is an isomorphism, it follows that for any A -local , one has Hom* Such an
Hom*
is an A-equivalence.

Example 7 (1)**.** Arguments similar to those given before show that if is a homotopy invariant pointed sheaf, then for any simplicial pointed sheaf , one has
/ Hom* (/ Hom/ A(/ in particular, if is smooth over and if is the disjoint union of and of a base point, Hom*
A
A

2.3 Derived Categories Vs. Homotopy Categories
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For any topos , which for us will be the category of sheaves on some site , the pointed homotopy category as well as the derived category are
obtained by localization. For the derived category, one starts with the category of complexes of abelian sheaves. The subcategory of complexes living in homological degree is naturally equivalent, by the Dold Puppe construction, to the category of simplicial sheaves of abelian groups. The equivalence is
A
E
) simplicial /
complex
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!LB> >
We will write for the inverse equivalence. For a point, and an abelian group,
is indeed the Eilenberg–Maclane space . For a complex not assumed to live in homological degree , we define
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= G = where G = is the subcomplex in = of the form =
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=
!LB-
= G = is right adjoint to the inclusion functor complexes in homological degree " all complexes so that) form a pair of adjoint functors: simplicial abelian sheaves complexes of sheaves

Theorem 8 (2)**.** *Assume that Spec! with ! perfect. Then, for / a presheaf with transfers, and as above, and , Hom* / , H . = / Note that*
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/ , /
In this theorem is the simplicial sheaf of abelian groups whose normalized chain complex is in homological degree . To prove the theorem we establish the chain of equalities,

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. = / Hom = / , Hom* = / , Hom* / , H*

A

(1.1)

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the first equality is proved right before Proposition 4, the second right after Proposition 5 and the last one follows from Lemma 1. be the forgetting functor from abelian sheaves to sheaves of sets. Its Let left adjoint is Z : the sheaf associated to the presheaf

ABL /

/ abelian group freely generated by /

In the pointed context , the adjoint is

/ Z& / Z/ Z We have the same adjunction for (pointed) simplicial objects.

Proposition 9 (2)**.** *On a site with enough points (and presumably always), one has*

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) /

Z from pointed simplicial sheaves to complexes of (1) The functor abelian groups transforms local equivalences into quasi-isomorphisms (2) The right adjoint transforms quasi-isomorphisms to local equivalences.

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ABL =

(! with the Nisnevich topology: for

/,LO> is a point, and they

The assumption "enough points' applies to any in and any point of ,

form a conservative system.

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Proof. Local equivalence (resp. quasi-isomorphism) can be checked point by point, and the two functors considered commute with passage to the fiber at a point. This reduces our proposition to the case when is just a point, i.e. to usual homotopy theory. In that case, (1) boils down to the fact that a weak equivalence induces an isomorphism on reduced homology, a theorem of Whitehead, and (1.1) reduces to the fact: for a complex , , computed using any base point, is . The are easy to compute because has the Kan property.

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Applying Proposition 8, we deduce from Proposition 2 the following. □

Proposition 10 (3)**.** *Under the same assumptions as in Proposition 2, for simplicial sheaf and a complex of abelian sheaves, one has*

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*Hom**

/ =

Hom?

) Z& / =

/ a pointed (3.1)

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Let be a sheaf and be deduced from by adjunction of a base point. We also write and for the corresponding "constant' simplicial sheaf. One has

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) Z& /) Z/ Z/ in degree zero For the pointed simplicial sheaf / , the group A(? Z/ = which now occurs at the right-hand side of (3.1) can be interpreted as hypercohomology of = "over / viewed as a space', i.e. in the topos of sheaves over / . For / defined by an object of the site , this is the same as hypercohomology of the site . As we do not want to assume = bounded below (in cohomological numbering), checking this

requires a little care. For a complex of sheaves over a site , not necessarily bounded below, H can be defined as the Hom group in the derived category . ? Z For in a topos and the topos : "F viewed as a space', besides the morphism

A(< </

of toposes </ < , i.e. the adjoint pair + + , we have for abelian sheaves an adjoint pair +' + , with +' and + both exact. By Proposition 8, +' + induce an adjoint pair for the corresponding derived categories. As +' Z Z/ , we get (3.2) Hom? Z/ = H </ + =

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hence

/ = H </ + = (3.3) Let us consider the particular case of (! with the Nisnevich topology. For any complex of sheaves, (3.3) gives for smooth over ! Hom = H %. = (3.4) Here, %. is the site (with the Nisnevich topology. It has however the same hypercohomology as the small Nisnevich site . . Indeed, one has a morphism F %. . and the functors F and F are exact. One again applies Proposition 8. If we apply (3.4) to a translate (shift) of = , we get (3.5) Hom* =, H . = Applying (3.5) to = / we get the first equality in (1.1).*

Proposition 11 (4)**.** *Let = be a complex of abelian sheaves on (! . The following Hom**

conditions are equivalent:

= '

1. is A -local. , the functor H is homotopy invariant. 2. For , one has in the derived category 3. For any complex in cohomological degree

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ZA = Hom =

Hom

Proof. By Proposition 3, condition (1) can be rewritten: for any pointed simplicial sheaf .

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) Z& / = Hom?) Z& / A A The operation / / A A is better written as a smash product / A

& / Z& Z& / . It follows that with A . For pointed sets and / , Z Hom?

Z&

/ A A Z& /

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Z& / Z& A Z& / ZA

(isomorphisms of simplicial sheaves), hence

) Z& / A A) Z& / ZA It follows that (1) is the particular case of (3) for of the form Z& . Similarly, (2) is the particular case of (3) for of the form Z , with . of a simplicial pointed sheaf is its smash product with The suspension the simplicial sphere (the -simplex modulo its boundary). It follows that

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9/

/ ' Z& 9 / Z& / Z&

/ '

(isomorphism of simplicial sheaves), and by Eilenberg–Zilber, the normalized complex Z is homotopic to the tensor product of the normalized complexes of Z& and Z& . The latter is simply Z :

) 9 / /

') Z& 9 / Z& / '

This is just a high-brown way of telling that the reduced homology of a suspension is just a shift of the reduced homology of the space one started with. Applying this to , one obtains that (1)(3). Indeed, Z& is & homotopic to Z Z . We now prove that (2)(1). For a complex, let (*) be the statement that the conclusion of (3) holds for all , . The assumption (2) is that (*) holds for reduced to Z in degree , and we will conclude that it holds for all in cohomological degree by "devissage':

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, ,

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, in degree zero, follows from Corollary 10. (a) The case of a sum of Z and that (*) holds (b) Suppose that is bounded, is in cohomological degree for all .

The functors

A(= A(ZA =

(3.6)

are contravariant cohomological functors, hence give rise to convergent spectral sequences

One has a morphism of spectral sequences

for for which is an isomorphism for , and both vanish for , J or , large. It

follows that for $, i.e. that satisfies (*). The same argument can be expressed as an induction on the number of ' such that

" . If $ is the largest (with $), the induction assumption applies to V 8 , even to V 8 , and one concludes by the long exact sequence defined by $ V 8 V

(c) Expressing as the inductive limit of the and using Proposition 11, one sees that we need not assume that is bounded. is a quasi-isomorphism, Z A Z A is one too (d) If (flatness of Z A), and (*) holds for if and only if it holds for . . For instance, (e) Any abelian sheaf is a quotient of a direct sum of sheaves Z the sum over , , of Z mapping to by . It follows that admits a resolution by such sheaves. By (a) and (c), satisfies (*). It follows from (d) that satisfies (*) and then by (c) that any complex in degree satisfies (*).

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□

2.3 Application to Presheaves with Transfers

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Let be a presheaf with transfers. A formal argument [7] shows that the presheaves with transfers are homotopy invariant. By the basic result (Theorem 1) recalled in the first lecture, it follows that for any , one has

= / H

= /

H

A = /

(4.1)

Indeed, as and A are of finite cohomological dimension, both sides are abutment of a convergent spectral sequence

= / H = / and the same for A . By Theorem 1 applied to = / , = / 7 = /

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is the same for or for Proposition 4((2)(1)) that

Proposition 12 (5). *For*

, is A-local.
A. Applying (3.5), we conclude from
! perfect, if / is a presheaf with transfers, for all „ = /
Combining Proposition 5 with Proposition 7 we get the second equality in (1.1).

2.5 End of the Proof of Theorem 2
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For any pointed simplicial sheaf , is a pointed bisimplicial sheaf of which one can take the diagonal . For any pointed sheaf , one has a natural map , and for a pointed simplicial sheaf , those maps for the induce
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0=

Proposition 13 (6)**.** *The morphism*
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0=
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0= is an A-equivalence.

Proof. We deduce Proposition 6 from Proposition 20.
The two maps
A are equalized by A , hence
become equal in the A -homotopy category. If two maps of pointed simplicial
sheaves factor as A , they also become equal. By
the adjunction of A and of A , such a factorization can be rewritten as
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/ = A(
/ = Particular case: the maps = , become equal in the homotopy category.
Evaluated on , these maps are the restriction maps A .
The affine space A is homotopic to a point in the sense that A A A interpolates between the identity map (for) and the constant map (for). The map induces
A A A and, composing with , in A , we obtain that A
A the identity map, and the map induced by A
A , are equal in the A -homotopy category. The map of simplicial sheaves is hence an A -equivalence. It has as inverse in the A -homotopy category the map induced by
=
Spec!
A and one applies Remark 1. We now apply Proposition 20 to the bisimplicial sheaves
= 0 . For fixed , this is just
A ,
and to the natural map and Proposition 20 gives Proposition 6.
To prove the last equality in (1.1), it suffices to show that:
/ , / , = / ,.

□

Lemma 14 (1)**.** *For any abelian sheaf lence from to*
/ ,

Proof. For

= / , induces an A-equiva-

a monoid (with unit), the pointed simplicial set

is given by

' functors from the ordered set $ viewed as a category) to

viewed as a category with one object

This construction can be sheafified, and can be applied termwise to a simplicial sheaf of monoids, leading to a pointed bisimplicial sheaf of which one can take the diagonal

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0=

This construction commutes with the construction . Indeed, is naturally isomorphic to , the operation commutes with products, and and

are both diagonals of the trisimplicial pointed sheaf . For abelian simplicial sheaves, the operation gives again abelian simplicial

sheaves, hence can be iterated, and . commutes with Via Dold–Puppe construction, corresponds, up to homotopy, to the shift of complexes:

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This can be viewed as an application of the Eilenberg–Zilber Theorem (see [9, Theorem 8.5.1]): one has

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Tot

(Eilenberg–Zilber),

and for each , the normalization of is just , so that the double complexes and for , otherwise, have homotopic . If is an abelian simplicial sheaf, applying Proposition 6 to , we obtain that

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0=

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<A

(6.1)

is an A -equivalence. The functor transforms chain homotopy equivalences into simplicial equivalences. For any simplicial abelian group (to be or), we hence have a simplicial homotopy equivalence

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Simplicial homotopy equivalences being A -equivalences, we conclude that (6.1) induces an A -equivalence

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)0= ,

□

2.6 Appendix: Localization

= be a category and be a set of morphisms of = . The localized category = is deduced from = by "formally inverting all & '. With this definition, it is clear that one has a natural functor MA = = , bijective on the set of objects, and that for any category E , / / Æ MA Hom= E Hom= E is a bijection from Hom= E to the set of functors from = to E transforming morphisms in into isomorphisms. Let

If one remembers that the categories form a 2-category, and if one agree with the principle that one should not try to define a category more precisely than up to equiv given alence (unique up to unique isomorphism), the universal property of above is doubly unsatisfactory. The easily checked and useful universal property is

the following: Æ is an equivalence from the category to the full subcategory of consisting of the functors which map to isomorphisms.

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/ MA A(= E

= A(= E /

in = is such that the functor * L& HomQ X = transforms maps in into bijections, then

Proposition 15 (7)**.** *If*

A(Q A(Q UK V

Proof. By Yoneda construction X , = embeds into the category = of con

travariant functors from = to Sets, while = embeds into = , identified by (a) with the full subcategory of = consisting of / transforming into bijections. For in = , with image in = , and for any / in = = , one has in =

X/

X$ / Indeed, by (a) and Yoneda lemma for = and = both sides are / . This means that X$ is the solution of the universal problem of mapping X into an object of = = . In particular, for as in (b), i.e. in = , X$ coincides with Hom

Hom

X , as claimed by (b).

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= < E<

□

Proposition 16 (8)**.** *Let be a pair of adjoint functors between categories and . Let and be sets of morphisms in and . Assume that maps to and and induced that maps to . Then the functors , between by and again form an adjoint pair.*

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Proof. The functors and induced by diagrams

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/ and

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are characterized by commutative

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@ E< = Adjunction can be expressed by the data of F PS and 2 S P- such that the compositions S S S S S are the identity automorphisms of S and respectively (see, e.g. [6]). S , By the universal property of localization, F induces a morphism F indeed, to define such a morphism amounts to defining a morphism loc S loc, S and S loc locS . Similarly, 2 induces 2 P- . The morphism S is induced by S , similarly for S S S S , O = E<

and the proposition follows.

□

Proposition 17 (9)**.** *Suppose that:*

= = Then, a coproduct in = is also a coproduct in = .

gives rise to a right calculus of fractions. 1. The localization 2. Coproducts exist in , and is stable by coproducts.

For the definition of "gives rise to a right calculus of fractions' see [10]. It implies

that for in , the category of with in is filtering, and that

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& & HomQ UK V colim - - HomQ

=

Proof. For in , let For the coproduct of

It is cofinal: for

be the filtering category of morphisms , %, one has a functor "coproduct':

&

in , one can construct a diagram

&

with

in .

in , and & dominates &. For any , it follows that HomQ UK V colim KI- HomQ

colim KI-+ Hom

colim KI- Hom

colimKI- Hom HomQ UK V meaning that is also the coproduct of the in = . +

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□

Corollary 18 (10)**.** *Suppose that in the abelian category arbitrary direct sums exist and are exact. Then, arbitrary direct sums exist in the derived category , and the localization functor*

E%

=% E% commutes with direct sums.

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E%

% =% %

Hom Hom is exact for abelian groups. Exactness of . in % ensures that a direct sum as of quasi-isomorphisms is again a quasi-isomorphism, and Proposition 9 applies to % and the set of quasi-isomorphisms, proving the corollary.

Proof. The functor factors through the category of complexes and maps up to homotopy. Direct sums in are also direct sums in . Indeed, HomG 2 . Hom .

If

% , ' is an inductive system of objects of %, the colimit of % is the cokernel .% = .%

colim

%

% % . If taking - is injective: it is the of the difference of the identity map and of the sum of the the inductive limit of a sequence is an exact functor, the map colimit of the

. %

. %

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each of which is injective, as its graded for the filtration by the . is the identity inclusion. Under the assumptions of Corollary 10 if a complex is the colimit of an inductive sequence , and if the sequence

. = .

(10.1)

is exact, then for any , the long exact sequence of cohomology reads

Hom

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Hom

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Hom

The kernel of is simply the projective limit of the Hom

. One concludes.

M'(

. The cokernel is

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□

Proposition 19 (11)**.** *Suppose that in countable direct sums exist and are exact. If the complex is the colimit of the , and if the sequence (10.1) is exact, for instance if either:*

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1. In inductive limits of sequences are exact.

2. In each degree , each , is the inclusion of a direct factor.

$

then, one has a short exact sequence

M'(Hom

Hom

M'(Hom

Proof. It remains to check that condition (2) implies the exactness of (10.1). This is

to be seen degree by degree. By assumption, the , have decompositions

compatible with the transition maps . . A corresponding decomposition of (10.1) in direct sum follows, and we are reduced to check exactness of the particular case (10.1) 2 of (10.1) when for " , i.e. when is a fixed from , and is otherwise. Let (10.1) be the sequence (10.1)Z in for Z. It is a split exact sequence of free abelian groups. Because direct sums exist, , for a free abelian group is defined and functorial in . It is a sum of copies of , indexed by a basis of , and is characterized by

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Hom%). The sequence (10.1)2 is (10.1) % and, (10.1) being split Hom

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(functorial in exact, it splits and in particular is exact.
Hom
V
V
of a complex The truncation is the subcomplex which coincides with in homological degree and is in homological degree . For any complex , one has colim
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5$
V
and this colimit satisfies condition (2) of Proposition 11. It follows that □

Corollary 20 (12)**.** *Under the assumptions of Corollary 10, for any short exact sequence*
M'(HomV
Hom
M'(HomV

3 A -Equivalences of Simplicial Sheaves on

2.4 Sheaves on a Site of

and , one has a
-Schemes
-Schemes
We fix a base scheme , supposed to be separated noetherian and of finite dimension; fiber product K will be written simply as . We also fix a group scheme over , supposed to be finite and flat. We note - the representable sheaf defined by . Let be the category of schemes quasi-projective over , given with an action of . Any in admits an open covering by affine open subschemes which are -stable. This makes it possible to define a reasonable quotient in the category of schemes over (rather than in the larger category of algebraic spaces). For each , is defined as the spectrum of the equalizer
[
[
O and is obtained by gluing the . It is a categorical quotient, i.e. the coequalizer of . The map is finite, open, and the topological space is the coequalizer of the map of topological spaces . One can show that is again quasi-projective. Remark 2 below shows that this fact, while convenient for the exposition, is irrelevant. One defines on [a pretopology [3, II.1.3] by taking as coverings the family of etale maps with the following property: admits a filtration by closed equivariant subschemes # such that for each + , some map has a section over . The Nisnevich topology on [is the topology generated by this pretopology. The category [with the Nisnevich topology is the Nisnevich site [. . O

Remark 21 (1)**.** The corresponding topos is not the classifying topos of [2, IV.2.5]. A morphism can become a Nisnevich covering after forgetting the action of , and not be a Nisnevich covering. Example: Spec , Z , , and permutes two copies of in .
!

Remark 22 (2)**.** Let affine . be the site defined as above, with "quasi-projective' replaced by "affine'. It is equivalent to . , in the sense that restriction to affine . is an equivalence from the category of sheaves on . to the category of sheaves on affine . .
[
[
L

Remark 23 (3)**.** If is the trivial group , the definition given above recovers the usual Nisnevich topology. For , the condition usually considered: "every point of is the image of a point with the same residue field of some ', is indeed equivalent to the condition imposed above. This is checked by noetherian induction: if a generic point of can be lifted to , some open neighborhood of can be lifted to , and one applies the induction hypothesis to)0= .

L

3

3

We write [. for the category of quasi-projective schemes over , with the

Nisnevich topology.

' P is a covering of in [. , there is a in [. whose pull-back to is finer than U. as in the definition of the Nisnevich

Proof. Fix a filtration # topology. We write for the quotient map . For in ,)0= is in some , by equivariance of the , and one of the maps

has an equivariant section & over . Let be the henselization of at . The map being finite, the pull-back of to is the coproduct of the N for . . The map from to being etale, the section &, restricted

to .)0= , extends uniquely to a section (automatically equivariant) of over

N N . Writing as the limit of etale neighborhoods of , one finds that has an etale neighborhood 6 such that has an equivariant section over -I: 6. The 6 form the required covering V.

We define the -local henselian schemes to be the schemes obtained in the following way. For in [, . a point of , and N the henselization of at . , take the fiber product -I: N . As is finite over , this fiber product is a finite disjoint union of local henselian schemes, and -local henselian schemes are simply the -equivariant finite disjoint unions of of local henselian schemes, for which is local. □

Lemma 24 (2)**.** *If U covering V of*

is -local henselian, the functor A(is a [. , i.e. it defines a morphism of the punctual site Sets [-I: N , the corresponding fiber functor is / AM'(/ -I: 6 , the colimit being taken over the etale neighborhoods of . in . The collection of fiber functors so obtained is conservative.

Proof. The functor A(commutes with finite limits. It follows from □

Lemma 2 that it transforms coverings into surjective families of maps, hence is a morphism of sites L& [. .

Proposition 25 (13)**.** *If point of the site to . . If*

To check that the resulting set of fiber functors is conservative, it suffices to check that a family of etale is a covering if for any -local henselian ,

A(A(

is onto. The proof, parallel to that of Lemma 2 is left to the reader.

3.2 The Brown–Gersten Closed Model Structure on Simplicial Sheaves on -Schemes We recall that a commutative square of simplicial sets (or pointed simplicial sets)

(13.1)

)

is homotopy cartesian (or a homotopy pull-back square) if, when

)

weakly equivalent to it and mapping to by (Kan) fibration: map from to is a weak equivalence.

.

/

[

Definition 26 (3)**.** A simplicial presheaf on . is flasque if tractible and if for any (upper) distinguished square:

is replaced by

) , the

/# is con-

%

(, − etale, + open embedding, , % and %), the square / /

/% /

is homotopy cartesian.

/ / 7/ [/ /

Theorem 27 (3)**.** *Let*

be a morphism of flasque simplicial presheaves.

If the induced morphism of simplicial sheaves is a local equivalence,

then, for any in , is a weak equivalence.

7/

/ [/ 7/

Proof. For a -scheme let . be the small Nisnevich site of and for a presheaf on let - be the restriction of to . . Our assumption that

> is a local equivalence implies that > is a local equivadefines a morphism of sites lence. The map . . and □

Lemma 2 implies that the direct image functor commutes with the associated sheaf functor and takes local equivalences to local equivalences. Therefore the mor phism > > > is a local equivalence. The presheaves

and are flasque on and by [8, Lemma 3.1.18] we conclude that . >

7/

7, / , /

7/ ,

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7/ ,

7, /

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/ , /> , /> /

is a weak equivalence.

In [1], Brown and Gersten define a simplicial closed model structure on the category of pointed simplicial sheaves on a Noetherian topological space of finite dimension. As in Jardine [11], the equivalences are the local equivalences . The homotopy category is hence the same as Joyal's, but the model structure is different: less cofibrations, more fibrations. The arguments of [1] work as well in the Nisnevich topology, for the big as well as for the small Nisnevich site, or for . , once Theorem 3 is available.

We review the basic definitions, working in be the sub. . Let

simplicial set of , union of all faces but the -th face. For , #. One takes as generating trivial cofibrations the maps of the form :

[

>0

!

[

;# Y - 0 - ;% for an open embedding, 0 >

Y -

\ %

Y $ Y ;

0 - ,

One then defines the fibrations to be the morphisms having the right lifting property with respect to generating trivial cofibrations (see, e.g. [5]), the (weak) equivalences to be the local equivalences, the trivial fibrations to be fibrations which are also (weak) equivalences, and the cofibrations to be the morphisms having the left lifting property with respect to trivial fibrations. Following [1] and using Theorem 3, one proves that the trivial fibrations can be equivalently described as morphisms having the right lifting property with respect to the following class of morphisms :

P

P# >0 - 0 - P% for open embedding, 0 > The maps of the form

^ %

>0 -

0 -

P are called generating cofibrations.

376 P. Deligne For and pointed simplicial sheaves, one defines a pointed simplicial set by A(0 Following [1], one sees that the classes of cofibrations, (weak) equivalences, fibrations, and are a simplicial closed model structure in the sense of []. This has the following consequences.

Corollary 28 (14)**.** *If is cofibrant and fibrant, for any pointed simplicial set has in the relevant homotopy categories*

, one

Hom In particular, taking ! 0 one gets Hom* Hom**

Corollary 29 (15)**.** *If is a cofibration and C a cofibrant object, then C C is a cofibration.*

Corollary 30 (16)**.** *If is cofibrant and is fibrant, then for any C Hom* C Hom Hom* C (16.1)*

In (16.1), Hom is the pointed simplicial sheaf with components the sheaves

of homomorphisms from to . We now apply this framework to prove the following criterion for A -locality.

/

0

[[

. If, as a simplicial

Proposition 31 (17)**.** *Let be a pointed simplicial sheaf on presheaf, is flasque, then is A -local if and only if, for any in ,*

/

/

/ / A is a weak equivalence. We recall that A -local means that for any category

one has the following in the homotopy

A(/ A(* A /*

(17.1)

Lemma 32 (3)**.** *A fibrant pointed simplicial sheaf is flasque.*

/

;

Proof. The right lifting property of relative the morphisms % means that for an open embedding, the morphism is a Kan fibration. As is a sheaf, an upper distinguished square

/

/

/

% gives rise to a Cartesian square

/ /

/% / As

/ / is a Kan fibration, this square is also homotopy Cartesian. / is flasque' is replaced by

□

Lemma 33 (4)**.** *Proposition 17 holds of the assumption " the assumption " is fibrant'.*

/

P

$

says that for any

Proof. "Only if' # for □

Corollary 14, for any pointed simplicial set , one has

, > is cofibrant. By

> / Hom* > / and > / is just /. If in (17.1) we take > , so that > A we get A Hom* / A Hom* / That this holds for any means that / / A becomes an isomorphism Hom*

in the homotopy category , hence is a weak equivalence. "If' We apply Corollary 16. As A is cofibrant and fibrant,

Hom* A /

/ Hom* Hom A /

and it suffice to show that

/

A /

Hom

is a local equivalence. This Hom is a simplicial sheaf claim follows.

/ A and the

/

/

be a fibrant

Proof. We can now finish the proof of Proposition 17. Let

replacement of . As and are flasque, is a weak equivalence for any . That all A be weak equivalences is hence equivalent

to all A be weak equivalences, while is A -local if and only

if is.

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/ / / /

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3.3 -Closed Classes The proof of the main theorem of this section will be postponed.

of morphisms of pointed simplicial sheaves is 0-closed if: 1. (Simplicial) homotopy equivalences are in . 2. If two of , and are in then so is the third. 3. is stable by finite coproducts. 4. If / is a morphism of pointed bisimplicial sheaves, and if all / are in , so is the diagonal 0/ 0 .

-closed if, in addition, it is stable by arbitrary coprodDefinition 5. The class is 0 ucts and colimits of sequences / with the property that, degree by degree, / / (resp.) is isomorphic to an embedding % % of pointed sheaves. -closure of the union of the □

Theorem 34 (4)**.** *The class of A -equivalences is the 0*

Definition 35 (4)**.** A class

classes of:

1. Local equivalences 2. Morphisms A

for in (! -closed. In particular, the class of A -equivalences is 0

3.4 The Class of A -Equivalences Is -Closed The properties 4(1), 4(2), 4(3) are clear. The last property is proved in Proposition 20.

%

Lemma 36 (5)**.** *Let be a pointed simplicial set and is fibrant and A -local, then 2 is A -local.*

Proof. Because

a pointed simplicial sheaf. If

is fibrant, for any , one has in the homotopy category

2 Hom*% Applying this to and A and using Hom*

(17.1)

% A % A one deduces from the A -locality of that of 2 .

□

Lemma 37 (6)**.** *Let be a morphism of pointed simplicial sheaves and % be a pointed simplicial set. If is an A -equivalence, then so is % % %.*

one has in the homotopy category Hom % Hom*% Replacing by a fibrant replacement, one may assume fibrant. Applying (17.1)*

Proof. One has to check that for any A -local

one is reduced to Lemma 5.

□

Lemma 38 (7)**.** *Let be a morphism of pointed simplicial sheaves. If and are cofibrant, then is a A -equivalence if and only if for any fibrant A -local , the morphism of simplicial sets*

is a weak equivalence.

Proof. "If' Taking

one deduces from the assumptions that

Hom* "Only if' The assumptions imply that and are fibrant. For any pointed simplicial set % one has % Hom* % Hom* Hom*

and similarly for

and one applies Lemma 6.

□

Proposition 39 (18)**.** *The coproduct of a family of A -equivalences*

is an A -equivalence.

Proof. There are commutative diagrams

+

+

where morphisms on the first line are cofibrations, and where the vertical maps are local equivalences, and similarly for . Replacing (resp.) by (resp.) we may and shall assume that the and are cofibrant. The coproducts , are then cofibrant too. One has

and similarly for the , and one applies Lemma 7, and the fact that a product of a family of weak equivalences of fibrant pointed simplicial sets is a weak equivalence.

□

Proposition 40 (19)**.** *The colimit*

colim/

colim

of an inductive sequence of A -equivalences equivalence.

/

/ is again an A-

Proof. One inductively constructs an inductive sequence of commutative squares

/

/

/

in which the vertical maps are local equivalences, the and are cofibrant and the transition maps

, are cofibrations. A colimit of local equivalences being a local equivalence, it is sufficient to prove the proposition for the sequence . We hence may and shall assume that

is a sequence of cofibrations and similarly for the . The

colimits and of those sequences are then cofibrant. is the limit of the sequence of If is fibrant and A -local, weak equivalences

/

/

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/ In the sequences and / the transition maps are fibrations of fibrant objects. It follows that the limit is again a weak equivalence: the of the limit map onto the limit of , with fibers M'(-torsors. It remains to apply

□

Lemma 41 (7)**.** */*

Proposition 42 (20)**.** *Let be a morphism of pointed bisimplicial sheaves. If*

all . are A -equivalences, so is

/

0/ 0 /

To prove Proposition 20 we will functorially attach to an inductive sequence of pointed simplicial sheaves , whose colimit maps to by a local equiv alence. We will then inductively prove that is an A -equivalence, and apply Proposition 19. We begin with preliminaries to the construction of the .

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Proposition 43 (21)**.** *Let be the category of finite ordered sets and increasing injective maps. For any category with finite coproducts,*

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the forgetting functor

\ 0 =*

0 =*

\ ,

has a left adjoint : "formally adding degenerate simplicies':

coproduct, over all and all increasing surjective maps of

& 0

**

?B 0 =

\ is the 0 , of copies

0 =

** * We define the wrapping functor as the composite . For the category of sets or of pointed sets one has the following.*

*\ *

=

Lemma 44 (8)**.** *The adjunction map*

?B

7 ? B is a weak equivalence.

=

Proof. We will prove it for the category of sets. The pointed case is similar. The fundamental groupoid of is the category with set of objects , in which all maps are isomorphisms, and universal for the property that:

V defines a morphism V > V > V. G , > G > G> G. One has ? B . To handle and it suffice to show that 7 induces an isomorphism of fundamental groupoids. For any and any , , & , is the identity of , . This results from (2) applied to & & , which gives & , & ,& , (1) (2) For

As generators of the fundamental groupoid, it hence suffices to take non degenerate . For relations, it then suffices to take those coming from non degenerate : the degenerate give nothing new. If we apply this to , we find as set of generators , and relations indexed by , the same relations as for . The functor commutes with passage to connected components and to passage to a covering. To handle higher , this reduces us to the case where (and hence) is connected and simply connected. In this case it suffices to check that induces an isomorphism in homology. It does because one has a commutative diagram

degeneracies

V G

G ? B

?B

? B 7

=

6

= ? B 7 = degeneracies

in which the first arrow is an isomorphism, the second the effect of and the composite is a homotopy equivalence.

$

&! &!

7 on homology,

□

Proposition 45 (22)**.** *For a pointed simplicial sheaf, let be the n-th skeleton of , i.e. simplicial subsheaf of for which*

is the union of the images of the degeneracies . One has push-out squares for

>0 &!? B

(22.1)

0 &! ? B Let now / be bisimplicial. Each / is simplicial, and they form a simplicial system of pointed simplicial sheaves. Let us apply ? B and &! to the first variable, i.e. to the simplicial sheaf / for each fixed (. We again have diagrams (22.1) and, taking the diagonal 0, one obtains push-out squares: / >0 0&!? B/

(22.2)

/ 0 0&! ? B/ where / now stands for the pointed simplicial sheaf / . This way the simplicial sheaf 0? B/ , which by Lemma 8 maps to 0/ by a local equivalence, appears as an inductive limit of (22.2).

Proof of Proposition 20: With the notations of 22 it suffice to show that the

0 &! ? B/ 0 &! ? B &! ? B $ $ 0 &! ? B/ $ $

(with applied in the first variable) are A -equivalences. We prove it by induction on . For , , and is assumed to be an A equivalence. From to , we have a morphism of push out squares

/

/

/

(22.2) for

(22.2) for

/ is an A-equivalence, by Lemma 6, so are its smash product with >0 and 0 . It remain to apply the As

Lemma 46 (9)**.** *Suppose given a morphism of push out squares*

1 2

3

1

2

3

4

4

which is an A -equivalence in positions 1 2 and 3. If in each square the first vertical map is injective, then the morphism of squares is an A -equivalence in position 4 as well.

Proof. Replacing the push out squares by push out squares of local equivalent objects, we may and shall assume that all objects considered are cofibrant, and that the vertical maps are cofibrant.

If is fibrant applying to each of the squares we get a morphism of cartesian squares, of pointed simplicial sets in which the vertical maps are cofibrations:

4 3

4 3

2 1

2 1 If is in addition A -local, it is a weak equivalence in positions 1 2 and 3,

hence also in position 4. By Lemma 7, this proves Lemma 9, finishing the proof of Proposition 20 as well as of the claim that the class of A -equivalences is -closed.

0

□

3.5 The Class of A -Equivalences as a -Closure In this section we finish the proof of Theorem 4.

Proposition 47 (23)**.** *The homotopy push-out of a diagram*

[is the push-out

(23.1)

A of

5

[

5 (23.2)

0 where the vertical map 0 0 0 is induced by > > 0

0 mapping 0 to (resp.) in 0 . In the case of simplicial sets, A maps to 0 with fibers above , above , and above . The homotopy push-out A is the diagonal of the bisimplicial object with columns 5 # 5 obtained by formally adding degeneracies to

5

0 0*

>

>

[

*in * (cf. 21) [maps to , maps to]. If a morphism of diagrams (23.1), the induced morphism from A to*

[is

A is hence in

the closure of the three components of diagonal (4(3), (4)). A commutative square

for the operation of finite coproduct and

induces a morphism

Example 48 (1)**.** Let

A

)

).

be a morphism. The homotopy push-out of

=

is the cylinder

(23.3)

.M of . The morphisms .M

are homotopy equivalences. To check that the composite cyl homotopic to the identity, one observes that cyl is the push-out of

is

cyl

0 > 0 0 0 0

0

(the vertical map induced by mapping to) and that the composite

is induced by , homotopic to the identity by a homotopy fixing . Similar arguments would show that the homotopy push out cyl of

.M

.M

is homotopic to

0

cyl

by

=

.

Example 49 (2)**.** In any category with finite coproducts, a coprojection is a map isomorphic to the natural map for some and . If in a push-out square of pointed simplicial sheaves

%

%

%

the morphism of the squares

) 5 %, the square (23.4) is the coproduct

is a coprojection:

=

%

=

(23.4)

and

(23.5)

%

)

is a homotopy equivalence, being the coprodand the resulting morphism A uct of the homotopy equivalences of Example 1 resulting from the two squares (23.5). The same conclusion applies if is a coprojection. A morphism of pointed simplicial sheaves is a termwise coprojection if each is a coprojection of pointed sheaves. Example: for any diagram

(23.1), the morphisms are termwise coprojections. For any morphism A , this applies in particular to .

.M

Proposition 50 (24)**.** *If in a cocartesian square (23.3) either a termwise coprojection, then the resulting morphism from closure of the empty set of morphisms.*

or A to

is

) is in the 0-

$

Proof. For each , we have a cocartesian square of pointed sheaves

[

)

Let us view it as a cocartesian square of pointed simplicial sheaves. By Example 2, it gives rise to a homotopy equivalence A . One concludes by observing that A is the diagonal of this simplicial system of morphisms.

)

)

□

Corollary 51 (25)**.** *If in a cocartesian square (23.3):*

)

or is a termwise coprojection, then: 1. is in the 0-closure of . 2. is in the 0-closure of .

Proof of (1): The morphism of cocartesian squares

[

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=

defines a commutative square

A

A

)

0 0

in which the vertical maps are in the -closure of the empty set by Proposition 24, while the first horizontal map is in the -closure of by 8.

Proof of (2): One similarly uses

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Proposition 52 (26)**.** *A pointed simplicial sheaf is reliably compact if it coincides with its -skeleton for some and each is compact in the sense that the functor commutes with filtering colimits. This property is stable by* *for a finite pointed simplicial set (finite number of non degenerate simplices) and implies that is compact.*

$ A(/

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Construction 27. Let

and) be classes of morphisms such that:

(a) Sources and targets are reliably compact. (b) Each in is a termwise coprojection.

)

from pointed simplicial sheaves to pointed simplicial P- such that:

We will construct a functor sheaves and a morphism (i) For any (ii) If

/, /

/ is in the 0 -closure of is in , the morphism

is a weak equivalence. is in , the morphism (27.1) is a Kan fibration. (iii) If

)

(27.1)

Let us factorize as cyl . As the second map is a homotopy equivalence, the first is in the -closure of . In the corresponding factorization of (27.1):

0

/ cyl / / the first map is a homotopy equivalence. To obtain (ii), it hence suffices that cyl be a weak equivalence. Replacing each in by the corresponding cyl , this reduces us to the case where

/ /

in is a termwise coprojection, and we will construct in this case a functor such that (iii) for in , (27.1) is a trivial fibration. (c) each

The conditions (ii), (iii) are lifting properties: for

in , in squares: >0 /

0 / for

in) , in squares: Y /

0 / In the first case, the data are morphisms 0 / and >0

agreeing on >0 , i.e. a morphism 0 and we want it to extend to by :

Y for in :

0

0

^ G

>0

/

. Similarly in the second case, with

>0

^ G

0

/

>0 replaced

/

(27.2)

for

in) : 0

Y

0
/
\ G
(27.3)
The left vertical maps are termwise coprojections, and their sources are compact. One now uses the standard trick of defining as the inductive limit of the iterates of functors , where is deduced from by push out, simultaneously for all
</
/ </
/
>0
/
in
Y
/
in
/
0
^ G
and
0
G
\
The push out is by
sources
0
)
a morphism which is a termwise coprojection. By 26, to check that the resulting is in the -closure of , it suffices to check that the left vertical morphism in (27.2) (resp. (27.3)) is in the -closure of (resp. of the empty set).
/
/
0
0
For (27.2), this is the map marked 3 in
>0 0
1
>0
3
0
2
The morphisms 1 and 3 Æ 2 are in the the 2 out of 3 property.
0-closure of . So is 2 by 26 and one applies
For (27.3), the diagram is
Y
Y
1
0
2
3
0

with 1 and 3 Æ 2 in the contractible.

0-closure of the empty set. Indeed, Y and 0 are both

Remark 53 (4)**.** Let be a property of pointed simplicial schemes stable by coproduct, and suppose that: (a) For (b) For

5%

) %

in , the and have property . in , is in degree isomorphic to the natural map for some having property .

$

The functor constructed in Construction 27 is then such that for any , each morphism where has is isomorphic to some

5 property . In particular, if the have property , so have the .

%

%

Proposition 54 (28)**.** *(Proof of Theorem 4) We apply Construction 11, on the site taking for and the following classes.*

)

: For any in the site, the morphism A

[,

(28.1)

and for any upper distinguished square

(28.2)

% the morphism

) : For any in the site,

A

(28.3)

#

(28.4)

/

)

, that (28.4) is in ensures If a pointed simplicial sheaf is of the form that each is Kan. That (28.3) is in ensures that for each upper distinguished square (28.2), the morphism

A is a weak equivalence. As each is Kan, A is the homotopy fiber product of % over , and % is homotopy cartesian:

is flasque.

, for each , A

Further, as (28.1) is in

is a weak equivalence: by Theorem 3, is A -local. Suppose now that is a A -equivalence. In the commutative diagram

/

/

/ -closure of the morphisms (28.1) and (28.2), the later the vertical maps are in the 0 being local equivalences. In particular, they are A -equivalences and / is an A-equivalence between A-local objects , hence is a local equivalence. -closure, proving Theorem 4. It follows that is in the required 0 The functor used introduced in 28 can also be used to prove the following lemma.

/

/

Lemma 55 (10)**.** *If is a filtering system of A -equivalences, then*

is again an A -equivalence.

AM'(/

/

Proof. Consider the square:

/

AM'(/

/ AM'(/

Since the functor commutes with filtering colimits, the bottom arrow is a filtering colimit of local equivalences, hence a local equivalence. The vertical maps are A -equivalences, hence the top map is an A -equivalence.

□

2.5 One More Characterization of Equivalences

[

Denote by the full subcategory in the category of pointed sheaves on generated by all coproducts of sheaves of the form - .

[

Theorem 56 (5)**.** *The class of local equivalences (resp. A -equivalences) in is the smallest class which contains morphisms A upper distinguished and has the following properties:*

[[

?

*0**

for

1. Simplicial homotopy equivalences (resp. and A -homotopy equivalences) are in .

?

, and are in ? then so is the third. / / is a filtering system of termwise coprojections in ? , then /

AM'(/ is again in ? . / is a morphism of bisimplicial objects, and if all / / are 4. If / in ? , so is the diagonal 0/ 0/ . 2. If two of 3. If

The proof is given in 32.

/ /

, / > , then is

is such that, for each

Lemma 57 (11)**.** *If the morphism is a weak equivalence, and if the and are all of the form in the -closure of the empty set.*

0

The proof will use the following construction.

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= =

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%>

= /

be a set of objects of , such that Construction 29. Let be a category, and let any isomorphism class has a representative in . Let be the functor which to a presheaf of pointed sets on attaches the family of pointed sets > Q . It has a left adjoint :

=

,

family

%> > Q

>

disjoint sum over the = and %> of > =

If is viewed as a category whose only morphisms are identities, the natural functor

’ = = defines a morphism of sites = = , both endowed with the trivial topology (any presheaf a sheaf), and ’ , ’ are the corresponding direct and inverse image

of pointed sheaves. By a general story valid for any pair of adjoint functors, for any pointed presheaf

on , the form a pointed simplicial presheaf augmented to :

/

/

=

’ ’ /

S/

7 S/ / Further:

/

’%

7

, i.e. of the form , is a homotopy equivalence. (a) If is of the form > is a homotopy equivalence: for each in , (b) For any , is a homotopy equivalence.

/

/ ’ 7

= S/

/ we define S/ 0simplicial system of S/

For a simplicial presheaf

392 P. Deligne

[

Proposition 58 (30)**.** *(Proof of Lemma 11) Let us say that in is connected if it is not empty and is not a disjoint union: should act transitively on the set of connected components of . Let*

be the full subcategory of connected objects. A sheaf on is determined by its restriction to . Indeed,

. To apply Construction 29, we will use this remark to identify the category of sheaves on to a full subcategory of the category of presheaves on . For any as in Construction 29, the functor takes values in sheaves, that is in the restriction of sheaves to . Indeed, for connected, is the > same, whether and are taken in the sheaf or in the presheaf sense.

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/ / = =

= [[=

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Fix / as in Lemma 11. For each $, the assumption on / ensures that S/ / is in the 0-closure of the empty set, and similarly for . For each (connected) , the morphism of pointed simplicial sets /

-closure of the empty set. It follows that is a weak equivalence, hence in the 0 ’ ’ / ’ ’ : the 5 over = of the > / >

-closure of the empty set. Iterating one finds the same for ’ ’ / is in the 0

’ ’ , and S/ S is in this 0 -closure too. It remains to apply the two out of three property to S/ S

/ / is a local equivalence and if the / and are all of for [upper , then is in the 0 -closure of the A

Lemma 59 (12)**.** *If the form > distinguished.*

/

Proof. We will use the Construction 27 for the class of morphisms A for upper distinguished, and for the class of morphisms . By Remark 4, if the are of the form , so are the > . In the commutative diagram

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/ the vertical maps are in the required 0-closure. They are in particular local equivalences and so is . One verifies as in 28 that / and are flasque. By Construction 27(ii), for each , is a weak equivalence, and it

remains to apply Lemma 11 to Ex(f).

/

□

Lemma 60 (13)**.** *If is an A -equivalence and if the the form , then is in the -closure of the A > distinguished and A for .*

0

[

Proof. Similar to the proof of Lemma 12.

/ and are all of for [upper

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S/

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□

Proposition 61 (31)**.** *Since for any simplicial sheaf the map is a local equivalence Lemmas 12 and 13 imply that for any local (resp. A -) equivalence , the morphism belongs to the -closure of the A for upper distinguished (resp. the A for upper distinguished and A for).*

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Proposition 62 (32)**.** *Proof of Theorem 5: We consider only the case of A -equivalences.*

Proposition 20 and Lemma 10 imply that A -equivalences contain the class . In view of Lemma 13 it remains to see that is -closed. The only condition to check is that it is closed under coproducts. Let

?

? 0

/ % be a family of morphisms in ? . For a finite subset P in % set aD D

2D

/

P ; we have a morphism aD aF and the map is isomorphic to the a colimD 2aD It remains to show that aD aD is in ? . This morphism is of the form P- / / where is in ? . Using the fact that ? is closed for diagonals we reduce to the case > . Using the same reasoning as above we further reduce to the case > . Consider the class of such that P- is in ? . This class clearly contains morphisms A , has the two out of three property and is closed under filtering colimits. It also contains simplicial homotopy equivalences. It contains morphisms of the form , A because such morphisms are For map

+

%

A -homotopy equivalences.

3 Solid Sheaves

3.1 Open Morphisms and Solid Morphisms of Sheaves

We fix and repeated in

as in Sect. 3.1, and will work in

(. .

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[. . The story could be

Definition 63 (6)**.** A morphism of sheaves is open if it is relatively representable by open embeddings, i.e. if for any morphism u (that is, -

, in [ ), the fiber product / : - mapping to - is isomorphic -stable open subset of . - for a In other words: should identify / with a subsheaf of and, for any & , there is open in and -stable such that the pull-back of & with respect to is in / if and only if maps to . u to >

The property "open' is stable under composition. It is also stable by pull-back: if in a cartesian square

/

(32.1)

u

/

is open, then if is open and u is an epimorphism, then is open. Indeed,

Lemma 64 (14)**.** *For / - a morphism, the property that / is represented by open in is local on (for the Nisnevich topology).*

Proof. Suppose that the cover , and that each / / - is represented by . For , /X / X is then represented by X with X the inverse image of . By descent for open embedding, the come from

some , we have locally on an isomorphism / > and by descent for

isomorphisms of sheaves one has /

>. Given a square of the form (32.1) with open and u an epimorphism, if & is in , & can locally be lifted to a section of . As is open, it follows that locally on , / : - is represented by an open subset. Applying Lemma 14 one concludes that is open. The same argument shows that if we have cartesian diagrams is open. This follows from transitivity of pull-backs. Conversely,

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with each

open and

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u+

/ u

onto, then

is open.

□

Proposition 65 (33)**.** *The property "open' is stable by push-outs.*

Proof. Suppose

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is a cocartesian diagram, with open. In particular is a monomorphism, and it follows that is a monomorphism and that the square is cartesian as well. The

is an is onto. The pull-back of by morphism

isomorphism (being a monomorphism) hence open. The pull-back of by is just , open by assumption. It follows that is open.

/

We now fix a class following stabilities

= of open embeddings

6 in [. We require the

6 are open embeddings and if 6 is in = , 6 6 is an open embedding in = , and if 6 6 is etale, with 2. If

for the complement of in 6 , then 6 is in = . The classes = we will have to consider are the following: 6 with 6 smooth. 1. The open embeddings 6 with 6 smooth such that the action of is free 2. The open embeddings on 6 . Equivalently: 6 is the union of and the open subset on which the Conditions 34. 1. If so is .

/

action of is free. 3. When working in

(: all open embeddings. /

= /

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□

Definition 66 (7)**.** A morphism is -solid if it is a composite where each is deduced by push-out from some

in . > -, is -solid. A sheaf is solid if # In the pointed context, a pointed sheaf is (pointed) -solid if the morphism is -solid.

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Example 67 (1)**.** For open in , let -I> be the sheaf - , with the subsheaf > contracted to a point . If is in , then -I> is solid: it is the push-out of > by . Thom spaces are of this form: starting from > a vector bundle on , one contracts, in the total space of this vector bundle, the complement of the zero section to a point.

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The class of solid morphisms is the smallest class closed by compositions and pushouts which contains all > in . By Proposition 33 solid - for morphisms are open. For a monomorphism of sheaves, define to be the pointed sheaf obtained by contracting to a point: one has a cocartesian square

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By transitivity of push-out, any cocartesian diagram

/

/ / . induces an isomorphism /

Proposition 68 (35)**.** *A morphism of sheaves / is = -solid if and only if it is a composite / / / / of monomorphisms where each // is isomorphic to some TI> T > for 6 in = . /*

is deduced by push-out from

Proof. If a morphism isomorphic to TI> . From this, "only if' results. Conversely, if we have

/

6 , / is

>

T

T I>

(35.1)

T I>

/

cocartesian, and if T is deduced by push-out from TI> lifts to , then > T . Indeed, the diagrams (35.1) being cartesian as well as cocartesian, we have a cartesian

>

T

/ If

is deduced from

T by push-out:

>

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/ /

T

/ 6

T

then and it follows that . Let us suppose only that we have v & etale, inducing an isomorphism from & v to and a lifting of T& . If & v , TI> to is an isomorphism. This is most easily checked by applying the fiber T I> T& I>& functors defined by a -local henselian : a morphism , if it does not map to , lifts uniquely to a morphism to & . The assumptions made hence imply that is a push-out of >& , T& . Note that by the second stability property of & & is in . We will reduce the proof of "if" to that case. We have to show that if a monomorphism is such that TI> with in , then is solid. The cocartesian square

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6 =

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(35.2)

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6 C I 6 ,

can induces an epimorphism TI> . The natural section of TI> on hence locally be lifted to or to : for some filtration #

of by closed equivariant subschemes, we have etale maps

with a (equivariant) section over , and a lifting of X or to TI> to . Note . We may:

C

6 6 C C 6 6 C , taking : here the lifting is to , 1. Start with 6 2. Assume 6 " 6 ; the succeeding liftings then cannot be to , : they must be to 3. Shrink , first so that it maps to 6 , next so that it induces an isomorphism from I 6 to C C As / is a monomorphism the cocartesian (35.2) is cartesian as well. The composition , T I> T I> TI> gives by pull-back a factorization of / as / /

C

with each

/

/

cartesian and cocartesian, hence / / T T . Further, the morphism I X , hence X T T I> lifts to T I> lifts to / . It follows that / / is a push-out of I6 , which is in = , and T I>

T

solidity follows.

/

□

Remark 69 (1)**.** Another formulation of Proposition 35 is: a morphism solid if and only if the pointed sheaf is an iterated extension of in , in the sense that there are morphisms

/

6 =

, of the form TI> .

Proposition 70 (36)**.** *If / is open and*

is

=-

TI> 's with

/

with each

is

=

= -solid, then is = -solid.

Proof. In the proof we say "solid' instead of " -solid'. Let (*) be the property of a sheaf that any open is solid. If is solid, sits at the end of a

/

chain #

with each

push out of some for in . We prove by induction on that > satisfies (*). For , is in . If - is representable and # is open, it is of the form > open in , hence solid by Condition 34(1). - for It remains to check that if in a cocartesian square

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(36.1)

the sheaf satisfies (*), so does . In (36.1), > - is a monomorphism and the square (36.1) hence cartesian as well as cocartesian. Fix open, and take the pull-back of (36.1) by . It is again a cartesian and cocartesian square and, being open, it is of the form

/

T

X

(36.2)

/ / where is open in and 6 . The diagram / /

/

/

/

expresses as an extension of -I > X by □

Remark 1 using (*) for and the fact that

X IT

-I>

-I > T

/ and one concludes by is in = .

Proposition 71 (37)**.** *The pull-back of a solid morphism by an open morphism solid. In particular, if is open and if is solid, so is .*

/

/

& is

Proof. Since the pull-back of an open morphism is open, it suffices to check the proposition for a push-out of > - for open in :

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-

Pulling back by , we obtain a cocartesian square

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-

/ / with open in and open in . This shows that / / is solid.

Suppose now that we are given two classes = and = of open embeddings satisfying conditions 34. We define = = as the smallest class stable by Condition 34 containing the

6 6 6 6 for 6 in = and 6 in = . □

Example 72 (2)**.** If = is a class of all open embeddings and = is the class of those 6 for which acts freely outside , then = = = .

Proposition 73 (38)**.** *If the pointed sheaves / and / are respectively = and = -solid, the / / is = = -solid.*

Proof. By assumption, / is an iterated extension in the sense of Remark 1 of pointed sheaves T I> with 6 in = . Similarly for / , with 6 in
= . The smash product / / is then an iterated extension of the T I> T I> T T I > T T > = = solid.

□

Definition 74 (8)**.** A morphism is called ind-solid relative to = if it is a filtering colimit of = -solid morphisms. Exercise 39. We take to be the trivial group. A section on of a push-out taken for instance in the lexicographical order, hence it is
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7
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6
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I / 6 a6 I 7 6
can be described as follows. For an open subset of and a section of on consider on the small Nisnevich site . of the presheaf which sends
to the set of morphisms such that and . A section of on is given by data: # T
7 ? I
N
6 I / 6
1. An open subset of . 2. A section of on .
' 7. a6 I on 6 where ' is the closed embedding and 7. denotes the associated Nisnevich sheaf. Exercise 40. In the notations of Exercise 39, if / is a sheaf for the etale topology, so is . For any , the 6 I as in (1),(2) above form a sheaf for the etale topology. It hence suffices to prove that for 6 I fixed, the datum (3) forms a sheaf for the etale topology. This is checked by using the following criterion to check if a Nisnevich
sheaf is etale. For . , and for a finite separable extension of !N , let OON
be deduced by "extension of the residue field' from the henselization ON of at
. . The criterion is that ,L /,LOON should be an etale sheaf on ,L!N 0 . Exercise 41. It follows from Exercises 39 and 40 that if / is ind solid, and if / is etale, then is etale. In particular, a solid sheaf, as well as a pointed 3. A section of
6
solid sheaf, are etale sheaves.

Remark 75 (2)**.** The same formalism of open and solid morphisms holds in the site of all schemes of finite type over with the etale topology.

3.2 A Criterion for Preservation of Local Equivalences

We work with pointed sheaves on following result
[. Our aim in this section is to prove the
a a /

Theorem 76 (6)**.** *Let be a functor from pointed sheaves to pointed sets. Suppose that commutes with all colimits, and that for any open embedding , is a monomorphism. Then if is a >*
local equivalence and if and are (pointed) ind-solid, then is a weak equivalence.
a
a
Suppose that
[
/ a

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[
is an upper distinguished square. Adding a base point, we obtain . The morphism is then a local equivalence. Let us check that is a A A weak equivalence. As commutes with coproducts, this morphism is deduced from the commutative square
a
a
a a X
a 2 a -
a
[
by applying the same construction (23.3). This square is cocartesian because is. The top horizontal line being a monomorphism, it is homotopy cocartesian, and the claim follows. As commutes with colimits, this special case implies that more generally one has
a

Lemma 77 (15)**.** *If is in the -closure of the a weak equivalence.*
0
A
as above, then a is
/ , S/ / is a [S/ / is / a local S/ S

Proposition 78 (42)**.** *(Proof of Theorem 6) For any pointed sheaf local equivalence. Indeed for any connected in , a weak equivalence by Construction 29. It follows that for equivalence,*
/
aS
is a commutative square of local equivalences. By Lemmas 15 and 12, is a weak equivalence. It remains to see that is a weak equivalence – and the same for . For this it suffices to see that for a pointed ind-solid sheaf , is a weak equivalence. As and commute with filtering colimits, the ind-solid reduces to solid, and by the inductive definition of solid, it suffices to prove the following lemma.
aS/
aS/ a/ a S
a/
/
be an open embedding. If in a cartesian square of pointed > -
[
/ / is such that aS/ a/ is a weak equivalence, the same holds for .

Lemma 79 (16)**.** *Let sheaves*

Proof. Consider the cocartesian square
[
S > S -
S/
S [
One can easily see that the top morphism is a monomorphism. It follows that is point by point homotopy cocartesian, and is a local equivalence. The functor of Construction 29 transforms a monomorphism into a coprojection of the form 5 . It follows that each is of the form > > and, by Lemmas 15 and 12, is a weak equivalence. It remains to show that is a weak equivalence.
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aS aS a
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S a aS/ a/ a[a [aS a aS a
Let us apply to the morphism of cocartesian squares . By Construction 29 both and are homotopy > > equivalences, and remain so by applying . We assumed to be a and weak equivalence. As are cocartesian with a top morphism which is a monomorphism (by the assumption on , for), it follows that is a weak equivalence. Hence so is .

□

4 Two Functors

5.1 The Functor
One has a natural morphism of sites
2 [.
[.
given by the functor
2 [[with the trivial -action Indeed, the functor 2 commutes with finite limits and transforms covering families
into covering families. In particular the functor presheaf
2 is continuous: if / is a sheaf on [. , the / with the trivial -action is a sheaf on [. . The functor 2 has a left adjoint Q . As 2 is continuous, the functor Q is cocontinuous, and the functor 2 from sheaves on [. to sheaves on [. is / sheaf associated to the presheaf /

Proposition 80 (43)**.** *The cocontinuous functor Q is also continuous, that is, if / is a sheaf on [. , the presheaf / on [. is a sheaf.*

Proof. By Lemma 2 it is sufficient to test the sheaf property of / for a covering of deduced by pull-back from a Nisnevich covering 6 of . Passage to quotient commutes with flat base change. Taking as base , this gives that -I: 6 6 identifies 6 with the quotient of -I: 6 by . Similarly, if 6 6 -I: 6 , the quotient by of the pull-back to of 6 is 6 again. This reduces the sheaf property of / , for the covering of by the -I: 6 , to the sheaf property of / for the covering 6 of .
Q /
Q
Q Q Q /
The functor
gives rise to a pair of adjoint functors between the categories of presheaves on and , with being
. As is continuous , it induces a similar pair of adjoint functors between the categories of sheaves. This pair is
[
[
2#
associated sheaf Æ Q 2 Q so that one has a sequence of adjunctions 2# 2 2 . If / on [is representable: / - , then 2# / -I: . In particular, 2# transforms the final sheaf K on [. , also called “point’, into the final sheaf on [. , and 2# 2 is a pair of adjoint functors in the category of pointed sheaves as well. It is clear that 2# takes solid sheaves to solid sheaves. We also have the following. □

Proposition 81 (44)**.** *Let / be a pointed sheaf solid with respect to open embeddings 6 of smooth schemes such that the action of on 6 is free outside . Then 2#/ is solid with respect to open embeddings of smooth schemes.*

Proof. If 6 is the open subset of 6 where the action of is free, then 6 6 and if 6 , a push-out of 6 is also a push-out of 6 : we gained that the action is free everywhere. The next step is applying 2# , from pointed sheaves on [to pointed sheaves on [. This functor is a left adjoint, hence respects colimits and in particular push-outs. It transforms > to >I: , and in particular, for , the final object into the final object. To check that it respects solidity it is hence sufficient to apply:

smooth over , then is smooth. Proof. If is finite etale, for instance , the case which most interests us, this is clear, resulting from being etale. In general one proceeds as follows. The assumption that acts freely on implies that is a -torsor over . In particular, is faithfully flat. As is flat over , this forces to be flat over . To check smoothness of over it is hence sufficient to check it geometric fiber by geometric fiber. For & a geometric point of , smoothness of $ amounts to regularity. As $ is smooth over &, hence regular, and $ $ is faithfully flat, this is [4] (an application of Serre's cohomological criterion □

Lemma 82 (17)**.** *If*

acts freely on

for regularity).

2

Proposition 83 (45)**.** *The functor # respects local (resp. A -) equivalences between termwise ind-solid simplicial sheaves.*

/ / be a local equivalence between termwise ind-solid simpli[. To verify that 2# is a local equivalence it is sufficient to in [and the map

2#/ ,LO- 2#/ ,LO-

Proof. Let cial sheaves on check that for any

2# is a left adjoint, the functor

/ 2#/ ,LO- (45.1)

6 in [, 6 commutes with colimits. For an open embedding is again an open embedding and we can apply to (45.1) Theorem 6. / be an A-equivalence. Consider the square Let / is a weak equivalence of simplicial sets. Since

@ S/ S/

/ / By the first part of proposition 2# takes the vertical maps to local equivalences. Since 2# commutes with colimits, Lemma 13 implies that 2# S is in the 0 closure of the class which contains 2# A for [upper distinguished and 2# A for in [. By Theorem 4 it suffice to prove that

morphisms of these two types are A -equivalences. For morphisms of the first type it follows from the first half of the proposition. For the morphism of the second type

2

and 2# D= !

it follows from the fact that morphisms # A

A are mutually inverse A -homotopy equivalences.

Define L2# A

A (and similarly on AA) setting 2#S/ L2# /

S/

2

where is defined in Construction 29. Proposition 45 shows that L # is well defined and that for a termwise ind-solid one has L # # .

□

5.2 The Functor
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2 / 2 /
?
[
As in Sect. 3.1, we fix and . We also fix in which is finite and flat over . a presheaf on , we define] to be the internal hom object For Hom] . Its value on is K . If is a sheaf, so is] .
/ /
[/ / ? / /

Example 84 (6)**.** Take and ? deduced from the finite group acting on $ by permutations. In that case, if / is represented by , with a trivial action of , then /] is represented by , on which acts by permutation of the factors.

Remark 85 (1)**.** If / is representable (resp. and represented by smooth over), so is /] . More precisely, if / is represented by in [, consider the contravariant functor on of morphisms of schemes from ? to , that is the functor
A(> ? K K This functor is representable, represented by some quasi-projective over (resp. and smooth). This carries an obvious action @ of , and @ in [represents /] . Proof: by attaching to a morphism ? its graph, one maps the functor considered into the functor of finite subschemes of ? K , of the same rank as ? , that is the functor
' subschemes of ? finite and flat over ,) K
K
with the same rank as
? K over .
The later functor is represented by a quasi-projective scheme Hilb, by the theory of Hilbert schemes. The condition that K be the graph of a morphism from to is an open condition. This means: let K K be a subscheme finite and flat over . There is open in such that for any base change , the pull-back T of is the graph of some -morphism from K to K if and only if maps to . This gives the existence of the required , and that it is open in Hilb. If is smooth the smoothness of follows from the infinitesimal lifting criterion. The quasi-projectivity follows from that of Hilb. On the functors of is clear. For in represented, the action , one has
: ? : ?
6 : : 6 ? 6 ? 6
. . < [HomAZ I: < Hom: < Hom? Hom: < K ? HomAZ I: < K ? /] <
?
=
Let be a class of open embeddings in " -solid'.
=
[. . We will simply say "solid' for
/ is a solid sheaf on [. , so is /] . / is open, i.e. relatively representable by If a morphism of sheaves % open (equivariant) embeddings, there is a natural sequence of sheaves intermediate between %] and /] . In the case considered in Example 6, and for > -, they are represented by the open equivariant subschemes of consisting of those n-uples for which at least ! of the are in . The formal

Theorem 86 (7)**.** *If*
definition is as follows. A section of] over is a section of over K . Let be the] equivariant open subscheme of K on which is in . The sheaf is] the subsheaf of consisting of those such that all fibers over of are of degree at least . The condition that the fiber at be of degree is open in]]] , and it follows that the inclusion of is open. For , in
]] is simply . For large, it is .
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is deduced by push-out from an open map

Lemma 87 (18)**.** *Suppose that , so that we have a cocartesian square*
(45.2)
% /
!
Then, for each , the cartesian square
fiber product %
/ %]
%
]
(45.3)
/ %]
is cocartesian as well.
[. has enough points: as a consequence of Lemma 2, for , the functor / AM'(/ -I: 6 the limit being taken over the Nisnevich neighborhoods of in , is a point (a fiber functor). The class of all such functors is clearly conservative. Such a functor depends only on -I: , where is the henselization of at , and can be any equivariant -scheme which is a finite disjoint union of local henselian schemes essentially of finite type over , and for which is local. We call such a scheme -local henselian, and write / / for the corresponding

Proof. The site each in and
[
fiber functor. We will show that (45.3) becomes cocartesian after application of any of the fiber] functors defined above. It suffices to show that for any in ,
the fiber of (45.3) above is cocartesian in . This fiber is of the form
/
/
&
& / %
L
&

and such a square is cocartesian if and only if whenever or is empty, the other is reduced to one element. Here, we also know that is a injective. It hence suffice to check that if is empty, then is reduced to one element. Fix in] , and view it as a section of over K . Let K be

] the open equivariant subset where it is in . The assumption that be in

means that the degree of the fiber of at a closed point of is at least . By -equivariance of , this degree is independent of the chosen . We have to] show that if is not in , that is if this degree is exactly , then is the

] image of a unique element of .

&

/ % !

&

/ ? %

/ % % %

& ? & / % . . ! &

The scheme ? K is a disjoint union of -local henselian schemes ? K . By assumption, (45.2)? K is cocartesian, hence if & is not in % on ? K , then on ? K it comes from a unique & in . Let ? K be the union of those ? K on which & is in %, and ? K be the union of ? K on which it is not. That & is in / %] but not in / %] , means that ? K is of degree - ! over . On ? K , & is in %. On ? K , it comes from a unique & in . The section & & in % on ? K , & on ? K of % over ? K is a section of % %] on lifting &. It is the unique such lifting: any other lifting & , viewed as a section of % on ? K , can be in % at most on ? K , hence must be in % on the whole of ? K which has just the required degree over . This determines & uniquely on ? K , where it is in %, as well as on ? K , where it is the unique lifting of & to .

Proof of Theorem 7: The induction which works to prove Theorem 7 is the following. As is solid, it sits at the end of a sequence

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/ / [

where each . We prove is a push-out of some open embedding in] by induction on that for any , is solid. For , is representable, X

]] hence so is (1). A fortiori, is solid. For the induction step X X one applies the following lemma to X

X.

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□

Lemma 88 (19)**.** *Let*

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/] be a cocartesian square with open in . Suppose that for any C , / is [ ]] solid. Then / is solid.

Proof. As / is open by Proposition 33, the /] are defined. It suffices to prove that for each + , the open morphism /] /] is solid.

By Lemma 18, this morphism sits in a cartesian and cocartesian square U V fiber product /

/

UV /]
/
-
(45.4)
/]
] By assumption, is solid. It follows that] ☐

Proposition 37 to the open morphism -
/
]
/ /
] - is solid too (apply]). As is open, -
408 P. Deligne
so is , and by Proposition 36, of a solid map.
is solid. The map is then solid as a push-out
% ?
/ / / , / / / / , /

Example 89 (7)**.** It is not always true that if is a solid morphism, so is] . Take the trivial group and two points (i.e.). Then] . For any sheaf , the inclusion of in is solid (deduced by push-out from #), and applying] , we obtain . Pulling back by the natural map from to one of the summands (an open map), we see that if] is solid, so is .

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Corollary 90 (46)**.** *If /*

In particular, if
/
is open and solid, then is pointed solid, so is] .
] /]
/
]
is solid.
] is open and one applies Theorem 7 and Proposition 36.

We now define for pointed sheaves on [. a "smash' variant of the construction / /] . If we assume that the marked point , / is open, it is /] /] / ,] that is /] with / ,] contracted to the new base point. This definition can be repeated for any pointed sheaf if, for any monomorphism % / , we define/ %] to be the following subsheaf of /] : a section & of /] / ? is in / %] if for any non empty , there exists a commutative diagram ?

Proof. with non empty and & in % on .

☐

Example 91 (8)**.** Under the assumptions of Example 6, if is open in with comple] is

ment and if has the natural > , then - - [, where action of . In particular, if is the Thom space of a vector bundle over (that is, / with / X contracted to the base point), then] is the Thom space of the -equivariant vector bundle . on .

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Proposition 92 (47)**.** *If a pointed sheaf / is pointed solid, that is if , / is solid, then so is /] .*

Proof. As /] is solid, the open morphism / ,] /] is solid too (Proposi
] tion 36), and , / is deduced from it by push-out.
The definition of /] immediately implies the following: □

Lemma 93 (20)**.** *Let be a -local henselian scheme. Then * /] /?*
where
? are -local henselian schemes such that ?
?
/] respects local and A-equivalences.

Proof. Let / be a local equivalence. To check that /]] is a local equivalence it is enough to show that for any -local henselian , the map /] / ? ?]

□

Proposition 94 (48)**.** *The functor*
is a weak equivalence of simplicial sets. This follows from Lemma 20. Let be an A -equivalence. By the first part it is sufficient to show that]]] is an A -equivalence. We use the characterization of] commutes with filtering col A -equivalences given in Theorem 5. Since imits and preserves local equivalences it suffices to check that it takes A -homotopy equivalences to A -homotopy equivalences. This is seen using the natural map
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